

# Intelligent Content Retrieval for Crime Investigation using MLLM

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**Abstract**—Fast and accurate retrieval of relevant case records, crime reports, witness statements etc has become essential in crime investigations. Traditional retrieval methods like keyword searching, filtering and timeline searches have limitations in retrieving textural data based on their context, which results in missing many relevant crime related records. So a query based system that can retrieve text data based on its context is essential for text retrieval in the field of crime investigation. To overcome the limitations of the existing system, the proposed system uses a multimodal large language model ImageBind for retrieval of relevant crime related text data based on the query given by the investigator. The proposed system integrates ImageBind with Retrieval Augmented Generation (RAG) for efficient retrieval of text data. The proposed system was evaluated using precision@k and recall@k for different values of k. The proposed system was able to achieve the best performance when k=7. For k=7, the system was able to achieve a recall of 0.86.

**Index Terms**—Multimodal Large Language Model (MLLM), ImageBind, Retrieval Augmented Generation (RAG), crime investigation.

## I. INTRODUCTION

As crime rates increase, there has been an exponential growth in the volume of criminal records, investigation reports, witness statements, and other text documents maintained by law enforcement agencies. Investigators often need to search and retrieve relevant documents from huge volumes of data which is very important for accelerating investigations, uncovering patterns, and linking cases.

Existing text retrieval methods used in crime investigation systems have several drawbacks. Most systems use keyword searches that depend on exact keyword matches, which results in missing relevant documents that use synonyms and related terms. Boolean searches rely on exact matches and do not understand semantic meaning. This leads to incomplete results. Timeline-based searching restricts retrieval to fixed timeframes, making it easy to miss critical records that fall

outside the selected period, and it does not assess relevance based on content. Filtering-based searches mainly concentrate on filtering data based on the last modification, date and size of data. This fails to retrieve data based on the context. These limitations highlight the need for more intelligent, context-aware retrieval approaches that can better understand investigative queries and retrieve relevant information even when phrasing, terminology, or time periods vary.

The proposed system uses a Multimodal Large Language Model (MLLM) ImageBind for text retrieval to overcome the disadvantages of traditional search methods used in searching and retrieving documents that are important for crime investigations. Although MLLMs are designed to handle multiple modalities like text, image, video and audio, they also excel in understanding natural language queries as well as the context of text data with deep semantic comprehension. This makes them highly effective even for pure text retrieval tasks. MLLMs are pre-trained on vast amounts of diverse data, which enables them to understand paraphrases, synonyms, related terms, and context. Unlike keyword or Boolean searches, MLLMs are context-aware, which means they can grasp the investigator's intent and retrieve documents that align with the meaning behind the query. The proposed system also integrates MLLM with Retrieval-Augmented Generation (RAG), which can further enhance the retrieval performance. Instead of leveraging Large Language Models (LLM) for text retrieval, MLLM is used here because text data related to crime investigations may have visual or situational meaning. MLLMs are better at capturing context and understanding investigative terms that may have visual or situational meaning (like weapon types, crime scenes, or incident descriptions) and also give the scope to extend the retrieval to other data modalities in the future. So thus the ability of the proposed system for semantic understanding, context-awareness and flexible language

handling makes MLLM-powered text retrieval a powerful and intelligent alternative to the existing search techniques used in investigative systems.

## II. RELATED WORKS

Wayne et al. [1] provides a comprehensive overview of recent advancements in dense retrieval, which uses pretrained language models (PLMs) to generate semantic vector representations for effective text matching. It explores key aspects such as model architectures (like dual-encoders), training methods (including contrastive learning), indexing techniques, and real-world applications. Z. Li et al. [2] introduces a novel approach to improve model selection for text retrieval tasks by leveraging large language models (LLMs). Kim et al. [3] presents transformer-based methods focusing on case law retrieval and statute law entailment. The authors employed the Sentence-Transformer model all-mpnet-base-v2 to generate vector representations of legal case paragraphs. They computed cosine similarities between query and candidate cases, constructed similarity histograms, and trained a gradient boosting classifier to identify relevant cases.

Sansone et al. [4] provided a comprehensive survey of Legal Information Retrieval systems, categorizing approaches into NLP-based, ontology-driven, and deep learning models. They also discussed key open issues such as handling legal ambiguity, long document structures, and retrieving semantically relevant legal content. Haitao Li et al. [5] proposed SAILER, a structure-aware pre-trained language model designed specifically for legal case retrieval. The model captures the hierarchical and logical structure of legal documents by encoding the fact section and reconstructing the reasoning and decision parts, leading to improved retrieval performance across multiple legal datasets. Nguyen et al. [6] introduced attentive deep neural models including Attentive CNN and Paraformer to effectively retrieve long legal documents by focusing attention on the most relevant segments. Their bilingual, hierarchical approach outperformed non-neural baselines across English, Japanese, and Vietnamese datasets and achieved top recall and F2 scores on the COLIEE benchmark.

Ni et al. [16] proposed a dual-encoder architecture based on pretrained LLMs for dense text retrieval. Their work showed that large language models, when used as dual encoders, generalize better across retrieval tasks, significantly outperforming smaller models in zero-shot retrieval scenarios. Gao et al. [17] presented a retrieval model derived from LLMs that can be trained without supervised relevance labels. The model leverages contrastive learning using passages drawn from general corpora. Zhai et al. [18] explored how large language models (LLMs) are reshaping information retrieval (IR), highlighting both the transformative opportunities they present such as conversational search, content generation, and intelligent task assistance and the significant risks they pose, including hallucination and lack of real-time knowledge. They argue that while LLMs won't replace traditional search engines, future IR systems will likely integrate LLMs as interactive agents that leverage, rather than substitute, search capabilities.

Borgeaud et al. [19] introduced Retro, a retrieval-augmented LLM that achieves improved performance on language understanding and retrieval tasks by augmenting the LLM's text generation with external document retrieval. Zhao et al. [20] presents a comprehensive empirical study on enhancing Retrieval-Augmented Generation (RAG) systems specifically for financial applications. Their study offers valuable insights into task-specific RAG pipeline optimization, emphasizing the importance of aligning retrieval methods and generation strategies with domain requirements. Ghali et al. [21] explore the integration of Retrieval-Augmented Generation (RAG) with in-context learning and semantic search to improve domain-specific knowledge retrieval. The authors propose a hybrid framework that combines dense semantic search methods with prompt-based in-context learning to enable large language models (LLMs) to generate more accurate and contextually relevant responses. Siran Li et al. [23] conducts a comprehensive investigation into the design and optimization of Retrieval-Augmented Generation (RAG) systems. The authors systematically evaluate various components of RAG architectures, including retriever types (sparse vs. dense), reranking mechanisms, context formatting, and prompt strategies, across a range of benchmark datasets and downstream tasks. Cheng Ye et al. [24] investigates the integration of learning-to-rank (LTR) techniques within RAG architectures to improve information retrieval in electronic medical records (EMR) systems. The paper emphasizes the importance of context-aware ranking in medical settings, where retrieving the most pertinent patient information is critical for accurate clinical decision support.

P. Joshi et al. [7] proposes a Retrieval-Augmented Generation (RAG) pipeline tailored for multi-modal documents that include text, tables, and images. The authors design a robust system by integrating multiple specialized models: advanced NLP models for textual content, structured data models for interpreting tables, and computer vision techniques for analyzing visual elements. B. Tural et al. [8] explores the integration of Retrieval-Augmented Generation (RAG) techniques with Large Language Models (LLMs) to enhance the accuracy and contextual relevance of AI generated responses. The authors discuss methodologies for combining external knowledge retrieval mechanisms with LLMs, aiming to address challenges such as information hallucination and outdated knowledge in standalone models. The paper also highlights that the RAG architecture enhances NLP applications by dynamically leveraging knowledge acquired from large scale LLM training data for more effective information retrieval. Chen et al. [9] introduced MuRAG, a Multimodal Retrieval-Augmented Generator that enhances open domain QA by retrieving both images and it achieved state-of-the-art performance improving accuracy by 10% to 20% on both WebQA and MultimodalQA benchmarks by using a joint contrastive and generative pre-training strategy.

## III. DATASET

A custom dataset of 500 crime-related documents was manually created. All the documents are in plain text(.txt)

format. These documents were not real crime records, but rather assumed and synthetically created based on common crime scenarios. The dataset includes witness statements, short crime descriptions, crime event summaries, and brief case summaries. Each document was designed to reflect real-world language, containing crucial crime-related terminology, event descriptions, suspect details and locations. The dataset includes 125 witness statements, 125 short crime descriptions, 125 crime event summaries, and 125 case summaries, allowing for a well-balanced representation of different crime-related texts. The documents are relatively brief, maintaining a concise and direct format to capture essential details of each crime event. Each document varies in length, ranging from 50 to 150 words, with witness statements and short crime descriptions typically being 50–100 words, while crime event summaries and case summaries extend up to 150 words. The dataset ensures coverage of different crime scenarios like robbery, shooting, assault etc.

#### IV. PROPOSED METHODOLOGY

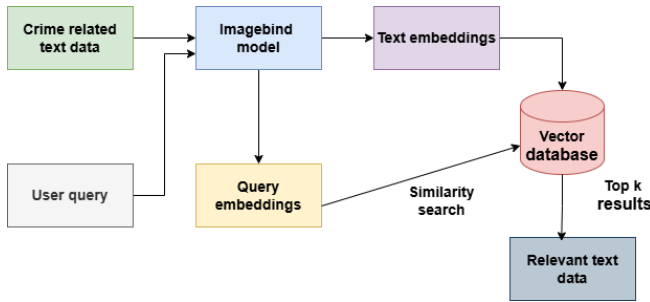


Fig. 1. Workflow of proposed system

We propose text retrieval for crime investigation by integrating multimodal large language model ImageBind with RAG. The goal is to overcome the challenges of the existing system and to enable investigators to retrieve relevant case related text data using natural language queries. Figure 1 shows the workflow of the proposed system.

ImageBind [10] is a multimodal model developed by Meta AI that can jointly understand and bind information from different modalities into a single shared embedding space. This means ImageBind can understand how different types of data relate to each other without requiring explicit paired data for training. ImageBind [10] was trained on large-scale datasets, covering a wide range of real-world situations, which gives it strong generalization capability, even in zero-shot settings. But here the proposed system uses ImageBind [10] only for text retrieval task as this will help to develop a richer semantic understanding of crime scene descriptions, case reports, and witness statements. For example, a text query like “armed suspect near a broken window” can be more accurately matched to crime reports that describe similar scenes, because ImageBind [10] understands not only the linguistic meaning but also the visual and spatial context

behind the terms. Thus it can work better than conventional LLMs for text retrieval.

As crime-related documents often contain important contextual clues scattered across different sections, these documents are first split into manageable chunks and are embedded using pretrained ImageBind model. This chunking strategy ensures that each retrieved segment retains meaningful context, while also allowing granular retrieval of only the most relevant portions instead of full documents. This approach works well for crime reports and case files, where each section of the document might address different aspects of the case such as incident description, suspect details, and other case related findings. These documents are often lengthy, unstructured, and written in multiple formats and languages. Therefore preprocessing of the text documents is done. The text is first cleaned by removing unnecessary special characters, multiple spaces, and irrelevant symbols. After cleaning, the text is tokenized and then broken down into smaller units (usually words or subwords) using a tokenizer compatible with the specific text encoder used within ImageBind [10]. In the case of ImageBind, the text encoder used is the same as that in CLIP[11], which uses a Byte Pair Encoding (BPE) tokenizer. Once tokenized, the text is converted into a sequence of token IDs, which are padded (if too short) or truncated (if too long) to fit a fixed sequence length expected by the model. These token IDs are then fed into the text encoder, which transforms the sequence into a dense vector embedding. Although ImageBind [10] is a multimodal model, here it is used purely for getting the text embeddings. Due to its multimodal pretraining, it captures a broader context than a standard text-based language model. This enables it to understand crime-related terminology and relationships between words in a more meaningful way. A linear projection head is used by the text encoder to create a d-dimensional embedding. This d-dimensional embedding is then normalized. A fixed dimension of 1024 is maintained for the text embeddings.

We adopt the retrieval component of the Retrieval Augmented Generation (RAG) framework to enhance the process of retrieving crime related text documents. The text embeddings obtained from the model are stored in the vector store. To insert our text data into the vector store, a table scheme is set up. This schema consists of file path and text embeddings. Investigators can input text query and this query is given to the ImageBind model. Investigators can enter natural language queries, which can be descriptive (e.g., “Burglary in an office with violent assault on staff.”) or concise (e.g., “weapons”). The system processes the query by embedding it using the ImageBind model, which maps it into a rich semantic space. The query is passed to the text encoder of the model followed by the linear projection head and after that normalized to a d-dimensional embedding of dimension 1024. Then a similarity search is done against the text query embeddings and the text data embeddings stored in the vector store to find the most relevant text data contextually similar to the user query. The similarity between the query and each document is computed

using cosine similarity calculation as shown in Equation 1.

$$\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} \quad (1)$$

Here A and B are text query embeddings and text document embeddings. A.B denotes dot product between the two embeddings. This measures how much the two vectors point in the same direction, which reflects their similarity.  $\|\mathbf{A}\|$  is the magnitude (length) of vector A.  $\|\mathbf{B}\|$  is the magnitude (length) of vector B.

This is the length of the document's embedding vector.  $\cos \theta$  is the cosine similarity score between the text query and the text document. The top-ranked documents with the highest similarity scores are returned to the investigator. Thus the proposed system provides a context-aware query based text data retrieval system for criminal investigations.

## V. RESULTS

To analyze the performance of the proposed system, we conducted quantitative and qualitative evaluations. The proposed system is compared with other models like RoBERTa and DistilBERT.

### A. Quantitative Analysis

Quantitative evaluation is done based on metric precision@k and recall@k for top k results.

Precision@K: It measures how many of the retrieved items are relevant. It is computed as in (2)

$$\text{Precision@K} = \frac{\text{Number of relevant items in top K}}{K} \quad (2)$$

Table 1 shows precision@k for text results when text query is given. Figure 2 shows a bar plot representation of the proposed system's precision@k compared with the other existing models. P@1 gives 1 for all models which means if one document is retrieved for a query then that is a relevant one. As k value increases the proposed system gives better precision than the other models like RoBERTa and DistilBERT.

Recall@K: It measures how many relevant items were successfully retrieved from the dataset. It is computed as in (3)

$$\text{Recall@K} = \frac{\text{Relevant items retrieved in top K}}{\text{Total relevant items in the dataset}} \quad (3)$$

Table 2 shows recall@k for text results when a text query is given. Figure 3 shows a bar plot representation of the proposed system's recall@k compared with the other existing models. As k value increases the proposed system gives better recall than the other models like RoBERTa and DistilBERT. R@k for k=7 give 0.86 for the proposed system than other models.

TABLE I  
COMPARISON TABLE FOR PRECISION@K FOR TEXT RETRIEVAL

| Model           | P@1 | P@3  | P@5  | P@7  |
|-----------------|-----|------|------|------|
| Imagebind + RAG | 1   | 0.66 | 0.80 | 0.85 |
| RoBERTa         | 1   | 0.66 | 0.60 | 0.71 |
| DistilBERT      | 1   | 0.33 | 0.40 | 0.51 |

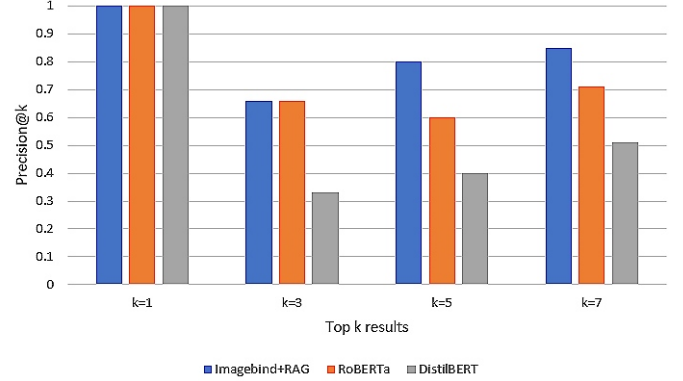


Fig. 2. Precision@k for text retrieval

TABLE II  
COMPARISON TABLE FOR RECALL@K FOR TEXT RETRIEVAL

| Model           | R@1  | R@3  | R@5  | R@7  |
|-----------------|------|------|------|------|
| Imagebind + RAG | 0.14 | 0.28 | 0.57 | 0.86 |
| RoBERTa         | 0.14 | 0.28 | 0.42 | 0.71 |
| DistilBERT      | 0.14 | 0.14 | 0.28 | 0.57 |

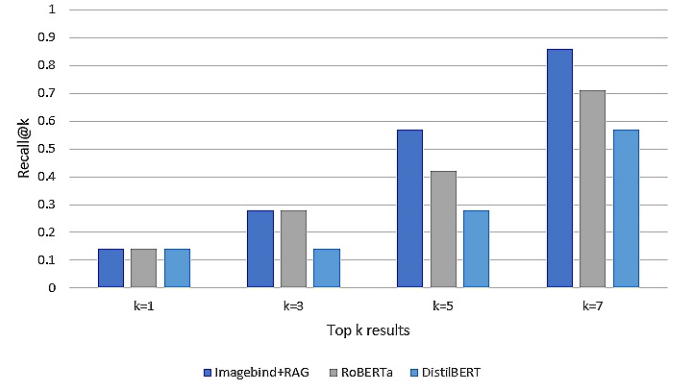


Fig. 3. Recall@k for text retrieval

### B. Qualitative Analysis

This section shows the snapshots of the retrieved text data related to crime. Figure 4 shows the retrieved text results when we give a query "Case summary related to the incident of a baby attacked by a woman". Figure 5 shows the retrieved text results when we give a query "Different case summaries related to physical assault". Figure 6 shows the retrieved results when we give a query "Witness statement about burglary".

25 The baby under the window showed signs of waking up and began to turn over. A woman in a pink top walked up to the baby and beat the baby with her fists. The woman stopped beating the baby after it stopped moving, covered the baby with a quilt, held the stroller, held the baby's hand, and then straightened the baby's quilt for a long time. The woman began to beat the baby.

Fig. 4. Query: Case summary related to the incident of a baby attacked by a woman

#### CRIME SCENE DESCRIPTION 1:

The baby under the window showed signs of waking up and began to turn over. A woman in a pink top walked up to the baby and beat the baby with her fists. The woman stopped beating the baby after it stopped moving, covered the baby with a quilt, held the stroller, held the baby's hand, and then straightened the baby's quilt for a long time. The woman began to beat the baby.

#### CRIME SCENE DESCRIPTION 2:

A woman with short hair, slightly fat, wearing a white top and black pants stood in front of the table and opened a book to read. A man wearing a white shirt and black pants entered the house and walked towards the short-haired and fat woman in front who was reading a book. The man suddenly pulled out a piece of red cloth from the left side of the woman's body then turned and ran away. When the woman turned back to the right he punched the woman in the head and ran out. The woman fell to the ground in pain and the book in her hand fell to the ground when she fell.

Fig. 5. Query: Different case summaries related to physical assault

25 My name is Arun Patel, and I would like to give a statement regarding what I saw. On the night of the incident, I noticed someone acting suspiciously near a house in my neighborhood. The person was dressed in dark clothing, wearing a hoodie and gloves. I saw them trying to force open a window at the side of the house. After a few minutes, they managed to get inside through the window. I couldn't see clearly what they were doing inside, but after about 15 to 20 minutes, the person came back out, carrying a backpack that looked heavier than when they first arrived. The person quickly walked away, looking around as if checking to see if anyone was watching. Although I couldn't see their face clearly, I could tell they were roughly medium height, average build, and appeared to be a young adult. I found the whole situation suspicious and felt it was important to report what I saw. I'm willing to help in any way if further information is needed. This statement is true to the best of my knowledge. Signed: Arun Patel

Fig. 6. Query: Witness statement about burglary

## VI. CONCLUSION

This paper shows the capability of using multimodal large language model ImageBind along with RAG for text retrieval in criminal investigations. The proposed system overcomes the problems of the traditional crime record retrieval systems which rely heavily on keyword matching, Boolean operators, or simple filtering, that often fail to retrieve text documents based on their context. By leveraging text embeddings using MLLM, the proposed system allows for semantic retrieval, ensuring that relevant case documents are identified even if the exact terms in the query do not match those in the records. The retrieval process is further enhanced by using RAG's efficient retrieval mechanism, ensuring both accuracy and fast retrieval while dealing with large crime document databases. The proposed system also achieves a recall of 0.86 when  $k=7$ . The proposed method successfully demonstrates how multimodal models, even when used in unimodal text-only retrieval tasks can significantly enhance the performance, paving the way for further research in exploring retrieval of new modalities of data.

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